1 June 2019

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 57 and 58

ACME Advanced Combustion via Microgravity Experiments
E-FIELD Flames Electric-Field Effects on Laminar Diffusion Flames

E-FIELD Flames is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The investigation was carried out in two rounds of tests, where the first (with ACME's coflow burner) occurred in Increment 55 and the second (with a gas-jet burner) began in Increment 56 and ended in Increment 57. As a result, there is nothing to be reported for Increment 58.

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I. Introduction

The Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment was conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this experiment, an electric field with voltages as high as 10,000 volts was established between the burner and a flat, circular mesh electrode suspended above (i.e., downstream of) the burner. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a strong electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Measurements were made of electric-field strength, the ion current passing through the flame, and flame characteristics such as the size, structure, temperature, soot, and stability. Conducting the tests in microgravity allowed for great simplifications in the analysis of ion-driven convection and flame interaction, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

An electric field can strongly influence flames because of its effect on the ions produced as a result of the combustion reactions. The induced ion wind can modify the flame shape, alter the soot and stability limits, and direct the heat transfer. The purpose of the E-FIELD Flames experiment is to gain an improved understanding of flame ion production and learn how the electric field can be used to manipulate the ions and control non-premixed flames for practical benefit. For example, electric fields can enable stable burner operation at fuel-lean conditions with reduced pollutant emission, i.e., where the flame is not reliably sustained without the electric field. As another example, the role of electric fields on soot formation can be dramatic. It appears that the ion driven wind alters the local convection and strongly influences the residence time of the soot precursors within the hot flame. The experiment contributes to our critical understanding of combustion processes in the presence of electric fields.

More specifically, E-FIELD Flames has the following experimental objectives:

A. Determine the relationship between electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and the amount of inert (nitrogen) in the fuel.
B. Obtain the time response of the flame chemiluminescence, luminosity, and the ion current to large step changes in voltage, as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
C. Evaluate and manipulate sooting behavior (i.e., soot volume fraction and location of luminous soot) in relation to the electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
D. Use the variation of the chemi-ion current as a function of applied voltage to characterize the combustion behavior of near-limit lifted flames and use the associated ion-driven wind to stabilize or reattach these flames.

The effect of the electric field on the flame is studied through tests conducted with a range of flame and electric field conditions. The fuel, fuel jet velocity, air flow, electric-field voltage and polarity were the primary parameters of the E-FIELD Flames experiment. Tests were conducted with methane and ethylene, including nitrogen dilutions of each, which provide a broad range of soot-producing conditions, i.e., from soot-free to highly sooting. They were carried out with both a simple gas-jet burner and the coflow burner that was also used for the Coflow Laminar Diffusion Flame (CLD Flame) experiment. Tests with the coflow burner were conducted with and without a coflow of air, where the results of the latter are compared with those from the gas-jet burner to assess the effect of air entrainment on the flame. The coflow burner also provided a different electrode configuration (close to a plane-to-plane) than the jet burner (close to a point-to-plane). In a typical test, the mesh electrode was energized at increasing potentials (relative to the burner), where the flame's response (e.g., ion current, luminosity, shape) to the resulting field strength was measured. The field strength was changed in repeated steps or was instead smoothly ramped.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the natural flicker of flames, which is a buoyancy-

driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Therefore, the flicker in the E-FIELD Flames was a result of electric body forces alone. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. For example, chemi-ion species, electronically excited chemiluminescent species, are at very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

E-FIELD Flames and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. For example, the mesh voltage can be adjusted to approach a flame's stability or soot limit. This was not done during Increments 55-56 or 57-58, where the console commanding during a test was typically limited to halting the test when it was observed in the analog ops video that the flame had extinguished.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 219 flames were ignited for the E-FIELD Flames experiment, with 142 and 77 respectively ignited during Increments 55-56 and 57-58. Note that the investigation's testing was completed during Increment 57, so no flames were ignited during Increment 58. The first and second rounds of testing respectively occurred in March-May 2018 and September-November 2018. The two rounds of tests resulted in 120 and 99 ignited flames conducted with coflow and gas-jet burners, respectively. The coflow burner was developed for the Coflow Laminar Diffusion Flame (CLD Flame) experiment, where CLD Flames' second round occurred between the two rounds of the E-FIELD Flames experiment. The 'interruption' in the E-FIELD Flames testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

E-FIELD Flames does not have living subjects like a life science experiment, but the tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH_4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C_2H_4), is highly sooting because of its carbon double bond. For both fuels, tests were conducted with pure fuel and fuel mixed with nitrogen over a broad range of dilutions. For limited cases, the fuel and nitrogen was premixed on the ground prior to launch. ACME bottles were launched with methane concentrations of 30%, 40%, 45%, and 100% with a balance of nitrogen (on a volume basis). Similarly, bottles were launched with ethylene concentrations of 30%, 50%, and 100%, again with a balance of nitrogen (on a volume basis). But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations. In all cases, the coflow was 21% oxygen with a balance of nitrogen (on a volume basis).

No samples (e.g., of soot) were collected for E-FIELD Flames; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the E-FIELD Flames ignited for E-FIELD Flames during Increments 55-57, including a limited overview of the test conditions, is found on the next page in Table 1.

Table 1.

E-FIELD Flames log of flames ignited in Increments 55-57 (*where the investigation was completed prior to Increment 58*)							
Test Day[1]	**Year**	**Month**	**Date**	**Mesh Polarity[2]**	**Fuel bottle**	**Fuel conc.[3]**	**Flames ignited[4]**
First round of tests, with the coflow burner, in March to May 2018 (Increment 55)							
18073	2018	3	14	negative	100% CH4	100%	10
18078			19			70%, 100%	10
18080			21		40% CH4	40%	8
18087			28		100% C2H4	80%, 100%	10
18092		4	2		30% C2H4	30%	11
18095			5	positive		30%	10
18102			12		40% CH4	40%	8
18106			16			40%	13
18113			23		100% CH4	100%	15
18130		5	10			70%	13
18137			17		100% C2H4	80%, 100%	12
Second round of tests, with the gas-jet burner, in Sept.-Nov. 2018 (Increments 56-57)							
18269	2018	9	26	negative	40% CH4	40%	11
18274		10	1		30% C2H4	30%	11
18282			9		100% CH4	70%, 100%	13
18285			12		100% C2H4	100%	10
18290			17			80%	11
18298			25	positive	40% CH4	40%	9
18302			29		30% C2H4	30%	10
18312		11	8		100% CH4	70%, 100%	13
18319			15		100% C2H4	25%, 50%, 80%, 100%	11

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Mesh Polarity** refers to the polarity of the high-voltage power supply and the copper electrode mesh relative to the burner.
[3]**Fuel conc.** (as diluted with nitrogen on a volume basis) was varied between tests, but typically matched the bottle. The list only accounts for the flames which ignited.
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

The CIR's IPSU-1 Image Processing and Storage Unit, which is used with the High Bit-Depth Multispectral (HiBMs) camera and the collimated illumination package, was only intermittently operable at the end of Increment 56 and was not operable at the start of Increment 57. Those capabilities were not seen as critical for E-FIELD Flames,

where the illumination is significantly blocked by the mounting bracket for the mesh electrode. Therefore, the tests on Sept. 26 (18269), Oct. 1 (18274), Oct. 9 (18282), Oct. 12 (18285), and Oct. 17 (18290) were conducted without HiBMs imaging or the CIR illumination. It was later determined to be an issue with a cable connection, which was corrected by astronaut Alex Gerst on Oct. 22 (18295) when he carried out other ACME tasks. As a result, the subsequent E-FIELD Flames tests were carried out using the HiBMs camera and illumination package.

2. Any other difficulties in obtaining scientific data

E-FIELD Flames tests sometimes automatically aborted (due to the controlling software script), where that was typically prior to gas flow and the activation of the igniter. As a result, there was little lost other than the time in the failed test attempt. The aborts normally occurred because of communication issues with IPSU-1. These are infrequent, but there are some days when tests aborted repeatedly.

Tests are of course also unsuccessful when the flame fails to ignite. The first test of the day has sometimes failed to ignite, presumably because the fuel line has not been sufficiently primed. Ignition was sometimes unsuccessful because the settings for the automatic flame detection are such that detection threshold is reached because of the igniter glow rather than a flame. As a result of the detection, the hot-wire igniter is deactivated and retracted, which is of course an issue if the flame has yet to ignite. This is typically avoided by detecting ignition with measurement of the flame's OH* chemiluminescence with a Photomultiplier Tube (PMT) module. The OH* chemiluminescence, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum. However, the combination of igniter settings and PMT gains can lead to false ignition detection.

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel, nitrogen, and synthetic air (i.e., 21% oxygen with a balance of nitrogen, on a volume basis) have unfortunately not always performed adequately. The fuel and especially nitrogen flows sometimes oscillate. Furthermore, the MFC used for the 'air' coflow behaves as if it is sticking in a few places, where the flow rate won't linearly ramp when commanded (as required for the CLD Flame experiment). It tends to stick in some places and then abruptly overshoot where that has sometimes been seen to extinguish the flame. As a result, mass flow controllers are tested - without flames - after they are first installed. Adjustments are often then made to (1) the settings that control when the valves open and (2) the Proportional-Integral-Derivative (PID) settings. This ACME activity has involved significant testing, separate from flame tests, and has taken place on 17333, 17334, 17347, 17354, 18008, 18022, 18038, 18046, 18143, 18241, 19059, and 19112. In this regard, there are an assortment of MFCs with different flow ranges for each of the three gases. While the MFC adjustments have been beneficial, they have not eliminated all gas flow problems. It was seen that nitrogen oscillations tend to damp with time, so a nitrogen pre-flow

was implemented so that the oscillations would typically end before the flame measurements. That approach was partially successful and was introduced during E-FIELD Flames' first round. Later, a procedure of re-zeroing the MFCs prior to each test was also implemented during E-FIELD Flames' second round. A replacement MFC for the 'air' coflow is being prepared for spaceflight, where a limited third round of CLD Flame testing is being planned to accomplish its extinction limit objective.

C. Completeness/Quality of Data

The number of flames ignited significantly exceeds the planned tests which were (1) specified as required in the ACME Integrated Science Requirements Document (ISRD), using the gas-jet burner, and (2) later expanded to include tests with the coflow burner to partially make up for science loss resulting from the slow response time of the high-voltage power supply. Following the revision, a total of 74 E-FIELD Flames tests were identified as required. In contrast, 219 flames were ignited in the experiment, so nearly three times as many flames were ignited than required.

It should be understood that a complete set of image data is not available for all tests. As discussed in section III.B, the IPSU-1 was not working at the beginning of the second round of testing, preventing use of both the HiBMs camera and the CIR's collimated illumination package. Those are not critical for E-FIELD Flames, so tests continued without their use.

Furthermore, recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done in order to maximize the scientific return of each day of testing within the project constraints.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). In this regard, note that the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS).

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and E-FIELD Flames is considered to be highly successful.

D. Status of Data Analysis

Data from the tests making up the E-FIELD Flames experiment includes video recordings from the operations camera for general flame behavior, high resolution images from the ACME data camera at various exposures and frame rates as dictated by the flame luminosity, intensified camera images, and non-imaging data at 125 Hz. The non-imaging data file is processed using Matlab scripts (modified from those developed by CLD Flame researchers) to separately show trends of ion current and flame luminosity as a function of the applied field, and the flame parameters (fuel, flowrate, burner type). The PMT data is also extracted to evaluate flame luminosity oscillations. Image data is processed using the OMA software (also developed and used by CLD Flame researchers).

The overall data volume is large with a storage limit of 127,000 MB per test day. All of the data has been downloaded to the hard drive with a backup system at UC Irvine. At this stage, the data has been sorted into appropriate segments so that comparisons can be made between burner geometries (jet versus coflow from the prior round of testing), fuel type and dilution (methane versus ethylene; no dilution, modest dilution, high dilution) and fuel flow velocity. The fuel flow velocity and dilution conditions were sometimes selected to hold total carbon flux into the flame constant. At this stage, the analysis remains comparative and qualitative, to show the major differences in behavior in microgravity and normal-gravity (1-g) flames, and to show particularly how buoyancy modifies the electrical response of the flames.

E. Preliminary Research Findings

Distinctly different results were seen in E-FIELD Flames' second round tests with the gas-jet burner when compared with the first round tests with the coflow burner. For example, the ion current response to the electric field strength with an upward ion driven wind in microgravity is a clear parabolic trend (which is similar to the trend observed in 1-g flames with the coflow burner as presented in the prior report). The parabolic behavior is evident even with the strong oscillations observed during the rise. This rising trend is not observed with the coflow burner.

The behavior of the soot and its charge state are complicated since in the regions of high field strength, particularly near the fuel tube where the field lines converge, soot can emit or collect free charges. Hence, the soot transport is driven directly by the field and by the ion-driven wind (and thermophoresis) which means it can travel differentially to the bulk flow. These various transport elements can push the soot through the flame to burnout, hold it in stasis in the flame for additional growth, or drive it to the burner where it accumulates. More work is needed to sort through these elements.

Positive electric field for 100% methane fuel with jet burner

The preliminary findings for the gas-jet burner in this section highlight the representative information of electric fields applied to a small hydrocarbon flame in the 0-g condition. The ISS tests covered a wide range of conditions which allowed the flame to settle as its

image was captured with the color camera and the intensified camera. The total flame luminosity and the ion current were also recorded in all cases. Figure 1 shows example flame images representative of the jet tests conducted with methane as fuel. The voltage is indicated as the nominal field strength (kV/cm). Since the physical spacing from the burner to the mesh is 35 mm, 1.16 kV/cm represents nominal potential of around 4000 V across the flame. It is important to note that with the sharp-edged jet burner, there is an electric field concentration near the burner tip, so the field in that region will be stronger than is indicated by the nominal field strength and stronger than would be seen for the coflow burner where the coflow portion of the burner acts as a planar electrode to mitigate the field concentration at the jet burner's tip.

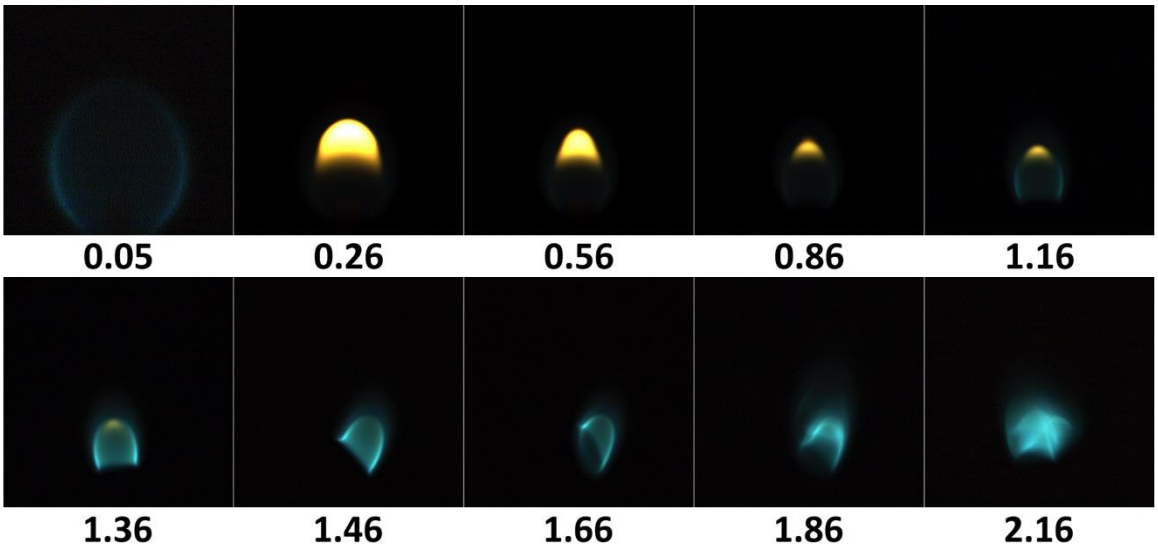


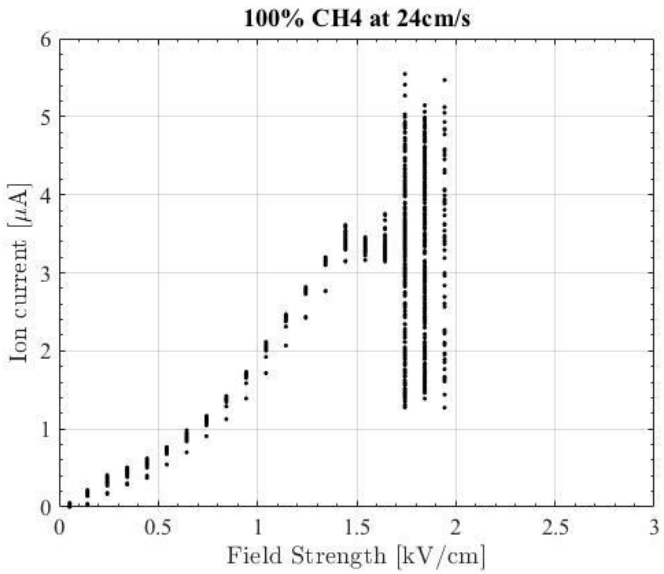
Figure 1. Flame images observed from the data camera at 19 cm/s methane jet with positive field applied. (Unit: kV/cm)

The very first image of each condition, e.g., at 0.05 kV/cm, demonstrates the initial voltage stage before the voltage sweep test commences. The jet flames are relatively more stable initially and less stable under high voltage than the previously reported flames using a coflow burner with no coflow [Chien et al., Ion Current and Flame Changes with Electric Fields in Microgravity, WSSCI Pasadena 2019]. As the positive field is applied to the jet, the flame becomes both narrower and shorter. An immediate change is the appearance of bright soot at the flame tip. With an increase of electric field strength, the soot region diminishes as the flame shrinks.

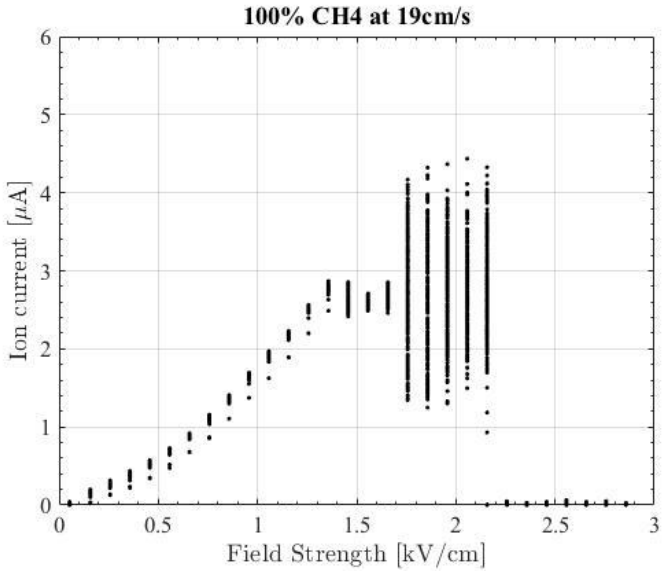
The voltage was increased 350 volts per step and held at each voltage for two seconds. The ion current data rate was 125 Hz and the monitored current fluctuation at the same voltage step is shown in the graph. The ion current to electric field strength (i.e., i-v) graph is plotted for 24 and 19 cm/s flow velocity in Figure 2, where the current initially increases parabolically with the applied voltage. The ion current corresponds to the flame appearance in that the current increases and reaches a peak at the most compact flame, e.g., at a field strength of 1.36 kV/cm for a 19 cm/s burner flow (as seen in Figure 2b). At

that maximum, the soot is almost gone (as seen in Figure 1). The flame base starts rising slightly around the current peak and lifts higher, with one side of the reaction wing attached to the tube, as the electric field increases further. The flame is not always lifted at the same edge of the burner tube; therefore, the attached reaction wing sometimes rotates along the tube edge with approximately the same lift angle under the same electric field strength. Otherwise, the lifted edge flame remains relatively stable. With further increases in the electric field strength, the flame starts fluctuating and a distinct dancing phenomenon can be seen from the i-v curve where the ion current oscillates strongly at the same field strength. The fluctuation crosses more than 3.5 μA (between ~1.2 to 5 μA) and reaches 5.0 μA for the 24 cm/s case. Similarly, this oscillation at the same voltage is observed for 19 cm/s with the ion current spanning around 2.5 μA (~1.4 to 4 μA) and reaching 4.0 μA; corresponding flame images are shown as 1.86 and 2.16 kV/cm in Figure 1. These very high peak values may represent an incipient corona or discharge event. The combustion extinguishes with the oscillations at 1.95 kV/cm for 24 cm/s and 2.16 kV/cm for 19 cm/s.

The flame geometries and soot distribution under the electric field influence clearly differ when comparing the gas-jet burner results with those from the earlier coflow burner tests without a coflow (i.e., with the same jet flow for each), where the annulus of the coflow burner acted as a planar electrode. But the maximum amount of ions produced within the flame (as evidenced by the ion current reaching the plateau) is similar for the case where the fuel flow velocity matches at 19 cm/s. The current also peaks around the same field strength per unit distance. Moreover, the oscillation behavior only exists in the jet burner experiment and furthermore influences the ion current monitoring due to the unstable flame behavior responding to the high field strength.



(a)



(b)

Figure 2. Ion current as a function of the field strength and fuel velocity. (a) 24 cm/s methane. (b) 19 cm/s methane.

Negative electric field for 100% methane fuel with jet burner

Unlike the positive field, the methane jet encounters the downward ion wind around the flame and the flame is not sustained after 1.86 kV/cm for 24 cm/s, being blown out by the electric field. The 19 cm/s flame lasts until 2.04 kV/cm. When the flame experiences a negative field, its height decreases and the full flame is moved onto the tube, similar to the downward pointing jet flame in 1g [Papac, M. et al., Canceling Buoyancy of Gaseous Fuel Flames in a Gravitational Environment Using an Ion-Driven Wind, Ann. N. Y. Acad. Sci, 1077, 2006]. The change of the flame width is not as obvious as with a positive field but it still changes, as revealed in Figure 3 for 0.05, 0.14, and 0.24 kV/cm.

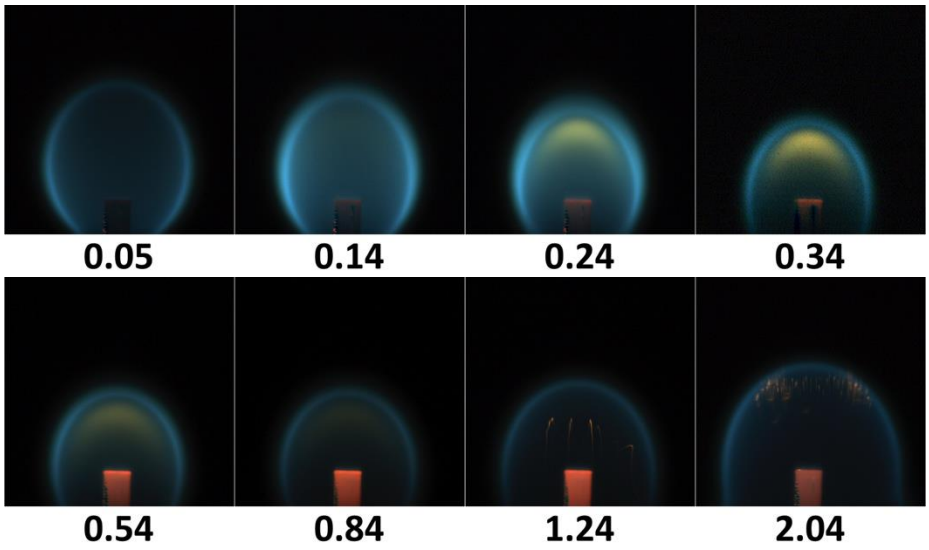
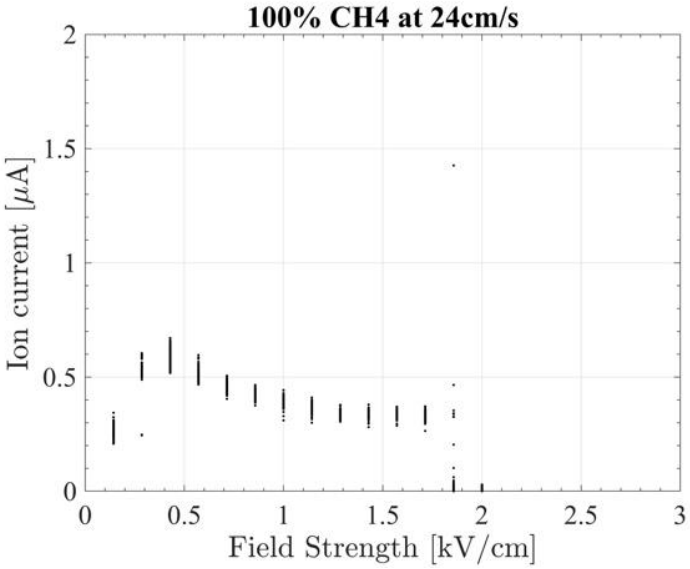
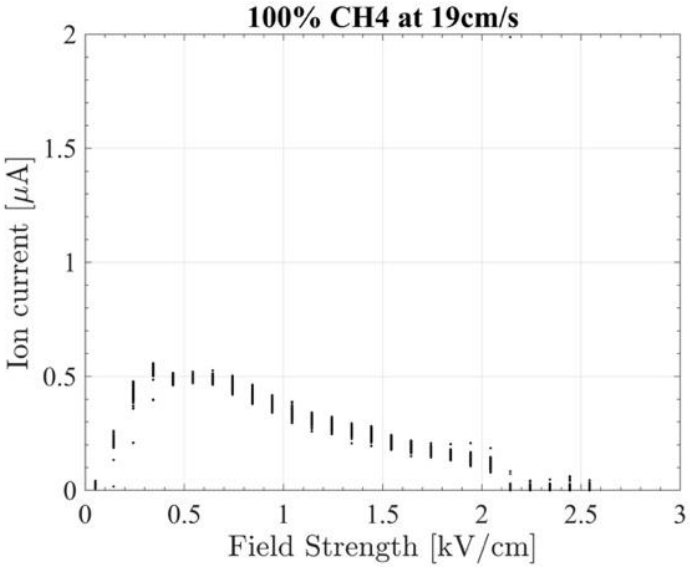


Figure 3. Flame appearance as a function of field strength observed from the data camera at 19 cm/s methane jet with negative field applied. (Unit: kV/cm)

The ion current (as seen in Figure 4's i-v curve) also increases with the field strength but differently than with the positive field at is has no clear curved slope. Instead, it decreases parabolically after reaching the current peak. The current peak corresponds to the most compact flame at 0.34 kV/cm for 19 cm/s and 0.43 kV/cm for 24 cm/s. The peak ion current is 0.53 and 0.55 μA, as shown in Figures 6b and 6a, respectively. Soot also appears at the flame tip but with a wider distribution along the reaction sheet. The soot reduces with increasing negative field strength, and it disappears at 0.84 kV/cm (as seen in Figure 3) along the parabolic decay line after passing the ion current peak. The flame width and height also expand after the most compact flame (at the peak current). This trend is demonstrated in the second-row of images in Figure 4. Notice that there are several sparkler-like particles, presumably of soot, at 1.24 and 2.04 kV/cm. Further evaluation of this sparking is underway.



(a)



(b)

Figure 4. Ion current as a function of the field strength and fuel velocity. (a) 24 cm/s methane. (b) 19 cm/s methane.

Positive electric field for 70% methane fuel with jet burner

The flame behavior with a lower hydrocarbon content of 70% methane diluted with nitrogen, shown in Figure 5, has a distinct reduction of soot at the flame tip in comparison to the cases with 100% methane (seen in Figure 1). The flame reaction wing starts lifting away from the burner edge when 1.06 kV/cm is applied, which is earlier than occurs in the 100% case, and the flame fluctuation also continues with increasing positive field strength.

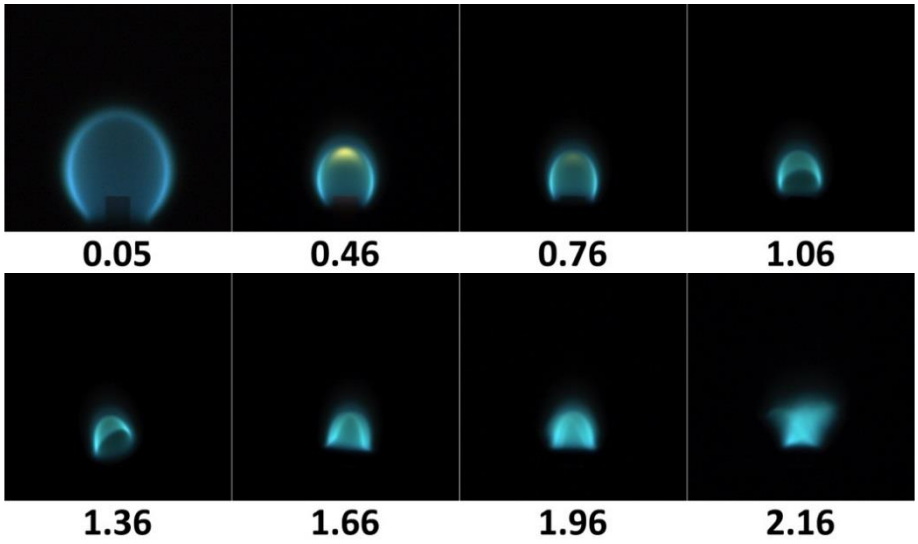


Figure 5. Flame appearance as a function of field strength observed from the data camera at 19 cm/s methane jet with negative field applied. (Unit: kV/cm)

The ion current to the electric field strength graph of 19 cm/s, shown in Figure 6, reveals a similar trend to Figure 2 for the 100% methane jet, and further details appeared in this condition. The ions are extracted from the flame gradually by the electric field, and with the increasing electric field the current rises. After reaching the ion current peak around 1.1 kV/cm, the flame has not yet oscillated as had occurred in the pure methane case. The ion current reduces and then rises back up in a parabolic behavior. The flame starts fluctuating around 1.75 kV/cm with the positive field strength increase. Further analysis is needed to understand the field strength to current behavior as a function of hydrocarbon content, and also with the influence from burner geometries in the microgravity environment.

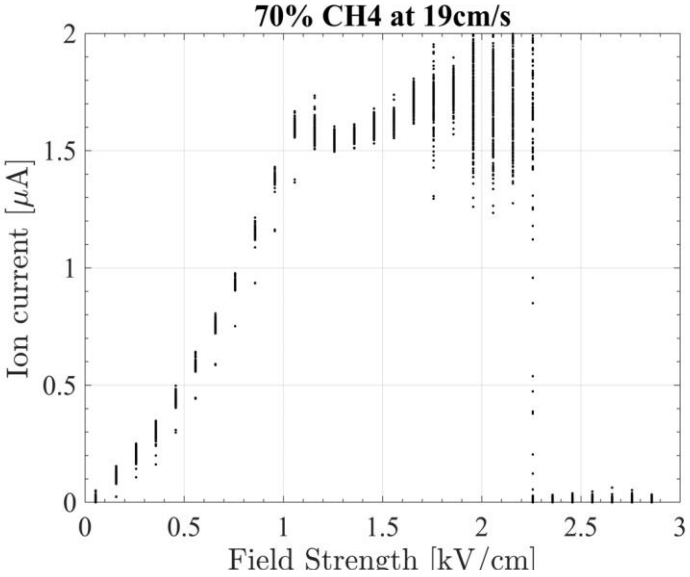


Figure 6. Ion current as a function of the field strength and fuel velocity at 19 cm/s methane.

In all of the i-v curves, the ion current data points at each field strength stretch across ~0.1 μA for each step, which are the jagged features from the spikes following a voltage step change that are mostly associated with the transient capacitance of the system when a rapid voltage step is requested. These overshoots notwithstanding, the i-v curves follow smooth and reproducible trajectories. The wide range of the ion current after the saturation plateau is due instead to the flame oscillation.

This report shows the initial results of the E-FIELD Flames experiment using the gas-jet burner in the International Space Station (ISS). The positive field allows the soot to concentrate and compresses the flame into a high current density thermal source. The flames demonstrate an unpredicted oscillation before extinguishing with increasing field strength. This oscillating phenomenon was not observed during the tests with the coflow burner from the first round, with the exception of a case that introduced the airflow at 30 cm/s which increased the surface reaction while the flame tended to balance between electric field effects, ions, and charged soot. The negative field allows the flame to maintain its spherical characteristic in microgravity. The yellow soot appears for both of the electric field directions within the flame around the tip and disappears with increasing field strength after reaching the current peak. The role of soot in the i-v behavior of these flames requires further investigation.

Early E-FIELD Flame results from the ISS testing were presented in following oral and poster presentations:

1. Derek Dunn-Rankin and Yu-Chien (Alice) Chien, “Electrical Field Effects on Laminar Diffusion Flames,” 11th Multi-Agency Coordinating Committee for Combustion Research (MACCCR) Annual Fuel and Combustion Research Review, Livermore, CA, 9-11 April 2018. (oral presentation)
2. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. “Electric Field Effects on Flames in Microgravity on the International Space Station,” Central States Section Combustion Institute Spring Technical Meeting, St. Paul, MN, 20-22 April 2018. (conference paper and oral presentation)
3. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. “Small Diffusion Flames under the influences of Electric-Field: Microgravity and Gravity,” 37th International Symposium on Combustion, Dublin, Ireland, 29 July to 3 Aug. 2018. (Work-in-Progress poster)
4. Yu-Chien (Alice) Chien, Jesse Tinajero, Dennis Stocker, Uday Hegde, Derek Dunn-Rankin, “Microgravity Experiments of Electric Field Effects on Laminar Ethylene/Air Diffusion Flames,” 34th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) Meeting, Rockville, MD, 31 Oct. to 3 Nov. 2018. (oral presentation)
5. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. “Ion Current and Flame Changes with Electric Fields in Microgravity,” 11th U.S. National Combustion Meeting, Pasadena, CA, 24-27 March 2019. (conference paper and oral presentation)

Upcoming presentations of ISS results include the following:

6. Chien, Y.-C., Stocker, D., Hegde, U. and Dunn-Rankin, D. "Microgravity Experiments with Methane Jet Flames under Electric Field Influence on-board the International Space Station (ISS)," 12th Asia-Pacific Conference on Combustion Meeting 2019, 1-5 July, Fukuoka, Japan. (final conference paper submitted)
7. Chien, Y.-C., Stocker, D., Hegde, U. and Dunn-Rankin, D. "Microgravity Experiments Examining Electric Field Effects on Laminar Methane Gas-Jet Diffusion Flames," 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) 20-23 Nov. 2019, Denver, Colorado. (abstract submitted)

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Videos describing the ACME project, its hardware, and the value of combustion research can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/.

Meanwhile, E-FIELD Flames videos can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/e-field-flames/

Videos related to E-FIELD Flames and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

None.